Math League Exam #1 Middle Grades
2024-2025 Governor's Cup Competition

1. The denominator of a fraction is greater than its numerator by 16. If negative 5 is added to both the numerator and the denominator, the new fraction equals 4/5. What is the original fraction?

a. 64/80
b. 69/85
c. 32/48
d. 74/90

2. Which of the following numbers is the greatest common factor for 10647 and 652509?

a. 39
b. 1521
c. 273
d. 59319

3. Find the value of X by using Vedic division to divide 66,922 by 9:

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        7 4 3 5  R 7
      ____________
9 √   6 6 9 2 2
        39
          3 2
            x 2
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a. 2
b. 3
c. 5
d. 7

4. A pattern includes the following numbers: 2/3, 1/2, 3/8, 9/32, and so on. Which of the following numbers is the next one in this pattern? The a^n term is $a^n = a*b^n$ and b = ¾.

a. 27/128
b. 9/64
c. 81/ 1024
d. 13/67

5. An arithmetic sequence has a first term of 14 and a fiftieth term of 161. Which of the following values is the sum of the first 50 terms of this sequence?

a. 7625
b. 4375
c. 5290
d. 8250

6. Given the following sequence, what is the missing value? The sequence is as follows: B2D4A1C3F6H8___.
   a. 5
   b. G
   c. E
   d. 7

7. In order to find three rational numbers between 2/7 and 3/7 that form an arithmetic sequence, which of the following formulas would you use to find the value to be added?
   a. (2/7 + 3/7) / (3-1)
   b. (2/7 – 3/7) / (3-1)
   c. (3/7 + 2/7) / (3+1)
   d. (3/7 – 2/7) / (3+1)

8. The sum of four integers may be represented by the following equation X + (X + 2) + (X + 4) + (X + 6). For the mean of these four numbers to be a positive integer, what is the lowest possible value for X?
   a. 0
   b. -1
   c. -2
   d. -3

9. What are the possible solutions for X so that $3|x - 2| = 21$.
   a. 9
   b. – 5
   c. – 5, 9
   d. – 9, 5

10. What is the solution of $C = \frac{a+b}{b-c}$ for a?
    a. $c + b$
    b. $c\frac{b+c}{b}$
    c. $b - c$
    d. c(b – c) – b

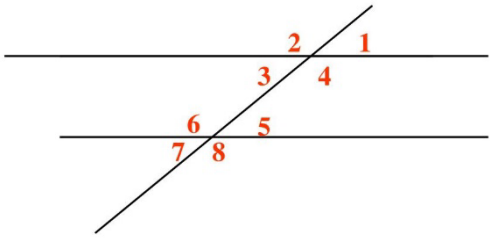
11. Alex purchased a baseball bat for $63.00. This was after the bat had been first reduced 30% and then later reduced by 40%. Which of the following is the original price of the bat?
    a. $120
    b. $150
    c. $175
    d. $190

12. A jar contains nickels and dimes in a ratio of 2:3. If there are 90 coins in the jar, then how much money is in the jar?
a. $7.20
b. $8.10
c. $6.30
d. $9.00

13. At the beginning of the year, the school store had spirit shorts and shirts in the ratio of 2:5. By the end of the first week of school, 20% of each type of spirit apparel was sold and 1400 apparel items remained unsold. How many shirts were there at the beginning of the week?
a. 900
b. 500
c. 250
d. 1250

14. Cynthia measured three pieces of string as 55 decimeters, 120 millimeters, and 160 centimeters. Which of the following is the equivalent measure of the combined lengths of the string?
a. 72 meters
b. 7.22 meters
c. 722 meters
d. 0.722 meters

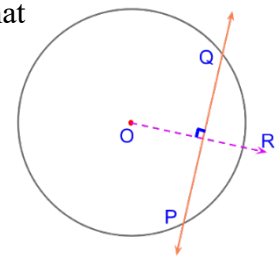
15. Which of the following measurements is the equivalent of one milliliter?
a. one cubic inch
b. one cubic decimeter
c. one cubic centimeter
d. one ounce



16. Two parallel lines are cut by a transversal. The measure of angle 2 is (4X – 10)° and angle 6 equals (11/3 X)°. What is the measure of angle 2?
a. 100°
b. 110°
c. 115°
d. 120°

17. In a scalene triangle, the lengths of two sides are 19 inches and 24 inches. BETWEEN what two integers must the length of the third side lie to create a triangle?
a. 4, 44
b. 19, 24
c. 5, 43
d. 20, 25

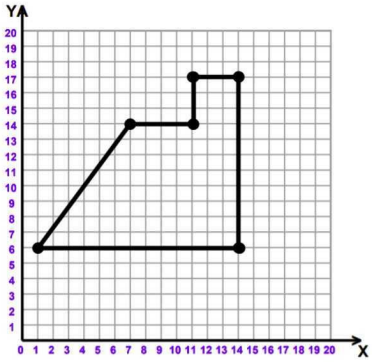
18. Circle O has a radius of 19 cm. The length of chord *QP* is 34 cm. What is the length of segment *rR*? Answers have been rounded to the nearest tenth of a centimeter.



a. 8.5
b. 10.5
c. 9.5
d. 19

19. A circle has a diameter of 69.12 centimeters. What is the area of this circle in square centimeters? Answers are rounded to the nearest hundred.
a. 1900
b. 4900
c. 1220
d. 3800

20. Pictured is an irregular shape drawn on the coordinate plane. In units, what is the perimeter of this shape?

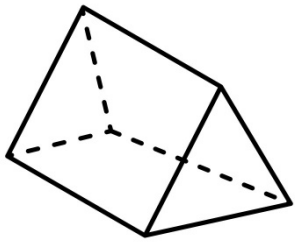


a. 44
b. 48
c. 43
d. 53

21. A trapezoid has bases measuring 19 inches and 26 inches. The height of the trapezoid is 15 inches. What is the area, in square inches, of this trapezoid?
a. 285
b. 337. 5
c. 390
d. 225.5

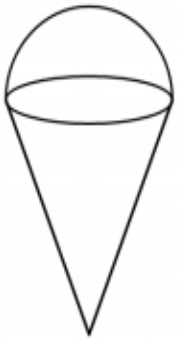
22. A rectangular prism has the following dimensions: 9 centimeters, 12 centimeters, and 20 centimeters. What is the surface area of the prism in square centimeters?
a. 840
b. 696
c. 1056
d. 2160

23. A triangular prism has a height of 13.6 inches. The base of the prism consist of a triangle with sides measuring 8 inches, 8 inches, and 9 inches. What is the volume of this prism in cubic inches? Answers are rounded to the nearest whole number.



a. 405 in^3
b. 367 in^3
c. 435 in^3
d. 377 in^3

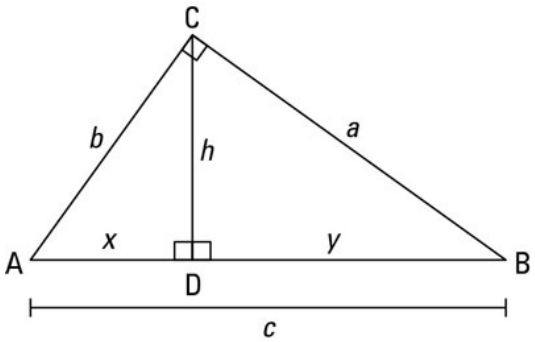
24. Pictured here is a cone with a hemisphere on top. The height of the cone is 4 inches. The radius of the cone and the hemisphere is 2 inches. What is the volume of the whole structure in cubic inches?



a. 40 /3
b. 25/3
c. 9 π
d. 32/3 π

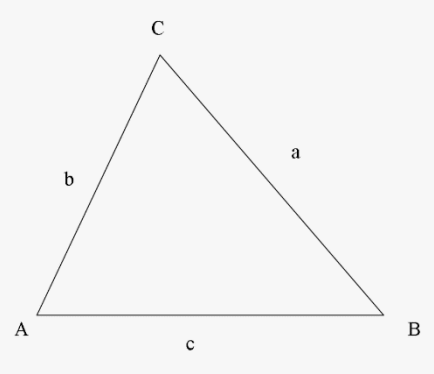
25. A square based pyramid has a height of 10 centimeters and a base measuring 25 square centimeters? In cubic centimeters, what is the volume of the pyramid?
a. 250
b. 75 1.3
c. 83 1/3
d. 100

26. For right triangle ABC, pictured here, x = 9 units and y = 16 units. Using what you know about similar right triangles, what is the length, in units, of altitude CD which is labeled *h*?



a. 15
b. 20
c. 12
d. 10

27. Angle A equals 58°, Angle B equals 48°, and Angle C = 74°. Which of the following proportions correctly uses the Law of Sines to find the length of side *c*?



a. sin 48/c = sin 74/b
b. sin 74/c = sin 58/b
c. sin 48/b = sin 74/c
d. sin 74/a = sin 58/a

28. A 30°- 60° right triangle has legs measuring 6 and $6\sqrt{3}$ units. What is the length of the triangle's hypotenuse in units?
a. 12
b. $12\sqrt{3}$
c. 18
d. $9\sqrt{3}$

29. Seasonal rainfall, in inches, for four Kentucky cities is shown below. Which city had the greatest percent of change in summer rainfall from 2023 to 2024?

City	Summer 2023	Summer 2024
Lexington	3.22	4.02
Pikeville	4.01	4.82
Paducah	4.23	5.13
Louisville	3.85	4.84

a. Lexington
b. Pikeveille
c. Paducah
d. Louisville

30. A data set consists of the following values: 88, 90, 86, 89, 96, 85, X . Which of the following values for X would cause the set to have a mean of exactly 90?
a. 94
b. 95
c. 96
d. 100

31. The shingles used on roofs have a mean of 20 years and a standard deviation of 3 years. If 2000 houses in a certain city are roofed every year, then how many of the houses will likely need a new roof in 14 or fewer years?
a. 50
b. 40
c. 80
d. 500

32. A spinner is divided into eight equal sectors: Red, Orange, Yellow, Green, Blue, Indigo, Violet, and White. A bag contains 8 blue marbles, 5 red marbles, 6 green marbles, and 4 white ones. What is the probablity that someone will draw a red marble and spin on red?
a. 5/184
b. 63/184
c. 3/92
d. 7/ 92

33. The two-way frequency table below shows the number of athletes competing in seasonal intramural sports who were injured last year and whether they exercised regularly throughout the year. What is the probability that an individual did not exercise and did not experience an injury?

	Injured	Not Injured	Totals
Exercised	240	900	1140
Did Not Exercise	260	20	280
Totals	500	920	1420

a. 46/71
b. 25/46
c. 1/23
d. 1/14

34. The school librarian has 12 sports biographies he wants to display. If he is able to display only four books at a time, then how many different ways can he display these biographies?
a. 495
b. 11,880
c. 5,940
d. 48

35. The graph of $Y \geq 1.5\,X + 3$ passes through which quadrants?
a. I, II, III
b. II, III, IV
c. II, III
d. IV, I, II

36. Given the following inequality, solve for X: $|(18 - 4x)| = 20 + 6x$.
a. – 1/5
b. -18
c. -1/5, - 18
d. -1/5, 18

37. When you graph the equation $y = {}^{3}/_{2}\,|x - 2| - 4$, what is the vertex for this graph?
a. (3, -4)
b. (-3, 4)
c. (2, -4)
d. (3/2, 4)

38. Which of the following expressions is the result when you mulitply $5 + \sqrt{3}$ by its conjugate.
a. 28
b. 22
c. 25 – 3i
d. 25 + 3i

39. A given expression $ax^2 + bx + c = y$ has two real solutions. The sum *of* $a + b + c = 14$, and $a = b + 3$. Which one of the following values for c is possible?
a. -1
b. 2
c. 3
d. 4

40. Solve for X: $\frac{2x+2}{3x-5} = 6$.
a. 3
b. no solution
c. 2
d. 5/3

41. What is the sum of the solutions for the quadratic equation: $3a^2 + 20a + 12 = 0$?
a. 20/3
b. 4
c. -20/3
d – 4

42. Sam takes 8 hours to repair a fence. When he works with Bill, it takes only 2 hours for them to complete the repairs. How long in hours would it take Bill if Bill worked alone?
a. 2
b. 8/3
c. 7/3
d. 4

43. The load a truck can carry varies directly with the cube of the diagonal of the truck's bed. A truckbed with a diagonal 8.75 feet will support a maximum load of 2500 pounds. In order for the truckbed to carry a load of 45,000 pounds, then how long in feet should the truckbed's diagonal be? Answers are rounded to the nearest whole number.
a. 20 feet
b. 157 feet
c. 23 feet
d. 125 feet

44. If x varies inversely with y and x = 45 when y equals 8, which of the following equations represents this relationship?
a. x = 360 y
b. y = 360 x
c. x = y/360
d. x = 360/y

45. For the function $f(x) = x^3 + x^2 \cdot x$, which of the following answers does NOT represent an equivalent value for F (4)?
a. 128
b. 2^7
c. 320
d. 10000000_2

46. Line *M* is represented by the equation $y = \frac{1}{2}x + 7$. Line *Q* is represented by the equation $-4x + 8y = 16$. When graphed, which of the following values is the point of intersection for lines *M* and *Q*?
a. No solution
b. (7, 16)
c. Infinitely many points
d. (1/2, 7/16)

47. The midpoint for line *PQ* is (8, -2a). If one endpoint is (-12, 2a), then what are the coordinates for the other endpoint?
a. (20, 4a)
b. (-32, 6a)
c. (-28, -4a)
d. (28, -6a)

48. Given $\sqrt{98} + \sqrt{72} + \sqrt{162} + \sqrt{50}$, which of the following is the most accurate simplification for this expression?
a. $22\sqrt{2} + \sqrt[3]{250}$
b. $22 + 5\sqrt{10}$
c. $\sqrt{332} + \sqrt[3]{250}$,
d. $27\sqrt{2}$

49. Match each of the following functions with the correct family of functions to which it belongs.

X. $f(x) = \sqrt{x+2}$	1. Square root
Y. $f(x) = 3^x$	2. Exponential
Z. $f(x) = \frac{4}{x-2} + 8$	3. Rational

a. X-1, Y-2, Z-3
b. X-1, Y-3, Z-2
c. X-2, Y-3, Z-1
d. X-3, Y-2, Z-1

50. Jamie throws a ball into the air from a height of 3 feet above the ground. The ball has an upward velocity of 50 feet per second. Gravity pulls the ball down, changing its position by about negative 16 feet per second squared. Which of the following indicates when the ball will hit the ground? Answers are rounded to the nearest hundredth.

a. 2.67
b. -.06
c. 3.18
d. 0.96

Governor's Cup League Test 1 2024-2025 KEY

1. B
2. B
3. C
4. A
5. B
6. C
7. D
8. C
9. C
10. D
11. B
12. A
13. D
14. B
15. C
16. B
17. C
18. B
19. D
20. A
21. B
22. C
23. A
24. D
25. C
26. C
27. C
28. A
29. D
30. C
31. A
32. A
33. D
34. A
35. A
36. A
37. C
38. B
39. A
40. C
41. C
42. B
43. C
44. D
45. C
46. A
47. D
48. D
49. A
50. C